

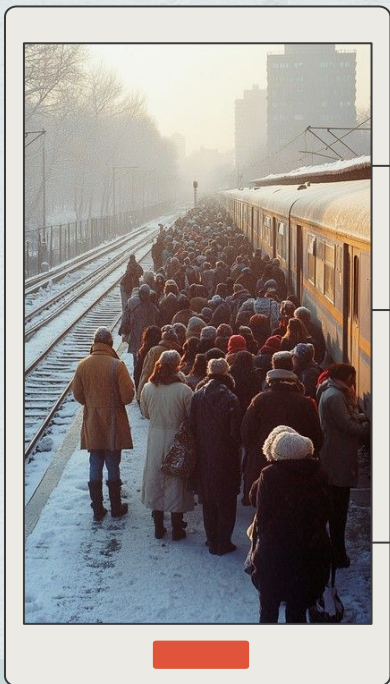
When Weather **Dera**ils The Schedule

Machine Learning for Finnish Train Delay Prediction

Team: Amanda Joseph and Winifred Paul



Project Overview



1. The Problem

Delays impact millions of passengers annually across Finland's rail network.

2. Our Objective

Evaluate whether machine learning models can reliably predict train delays using integrated weather and schedule data.

3. Research Question

To what extent can weather and schedule data reliably predict train delays, and where do model performance limits lie?



Data Selection



Finland Integrated Train-Weather (FI-TW)

- **Source:** Combined Finnish railway operations with FMI weather logs.
- **Scale:** Covers 5,900+ km of track supporting 90M+ yearly passenger journeys.
- **Conditions:** Merges transit metrics with extreme Arctic and Nordic coastal climate data.



How We Prepared the Data

From 125 raw columns to 14 modelling predictors

RAW DATA

-  6.28M records
-  125 columns
-  Train + weather data

TIME & DATE

-  Scheduled time → minutes
-  Month + day of week
-  Cyclical Encoding

WEATHER

-  Temperature
-  Wind speed
-  Snow depth
-  Visibility

 **Excluded: Precipitation (86% missing) | Station name (466 stations)**

FINAL FEATURE SET

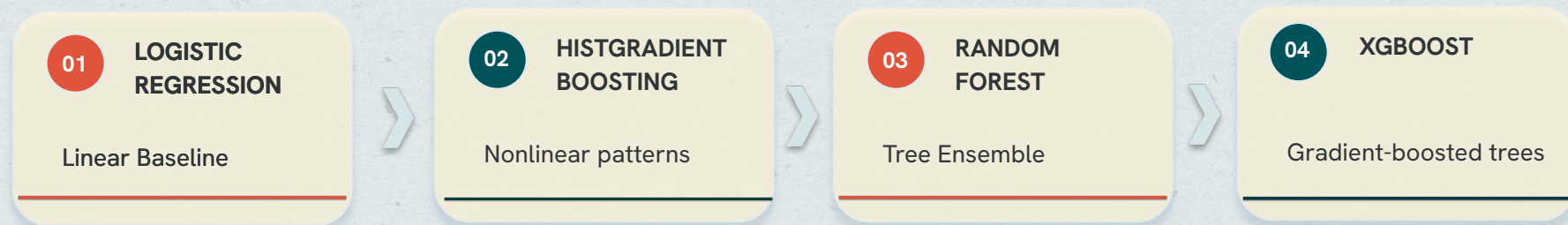
Train type + Arrival/Departure + Weather ▪ 14 predictors ▪ Target: Delayed / Non-Delayed



How We Tested the Models

A progression from a simple baseline to nonlinear tree-based model

MODEL PROGRESSION



HOW WE MEASURED



ROC-AUC

Primary Metrics



F1 Score

Secondary Metrics

HOW WE VALIDATED

3.48M

Training



3-FOLD

cross-validation



1.21M

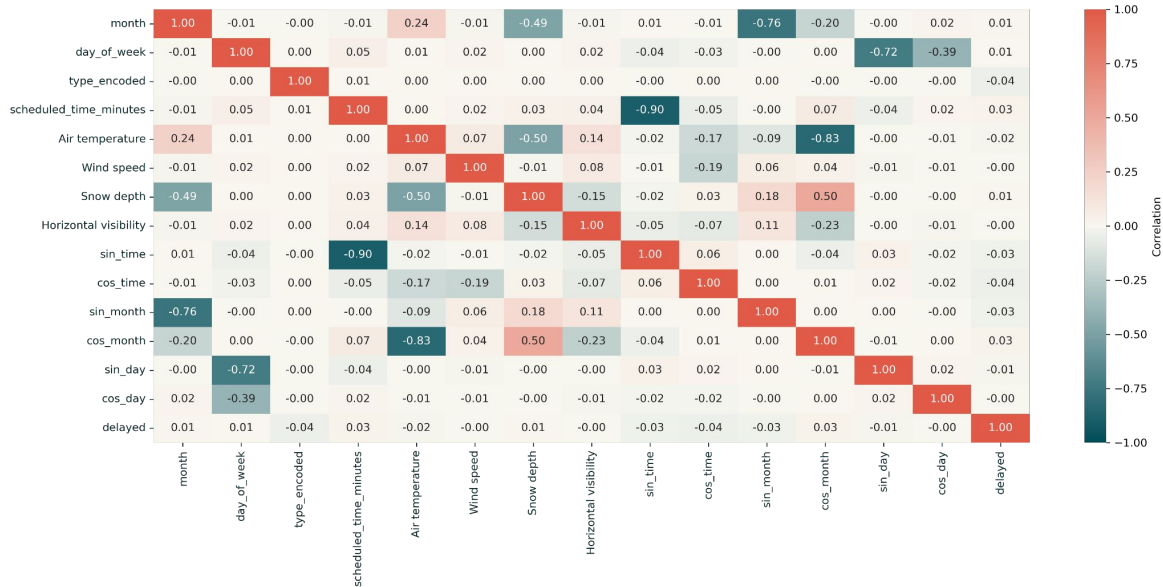
untouched test



From Straight Lines to Smart Decision Trees

Why simple correlations fail to predict complex delays

Correlation Heatmap - Training Data



01 LOGISTIC REGRESSION

ROC-AUC 0.55

F1 Score 0.60

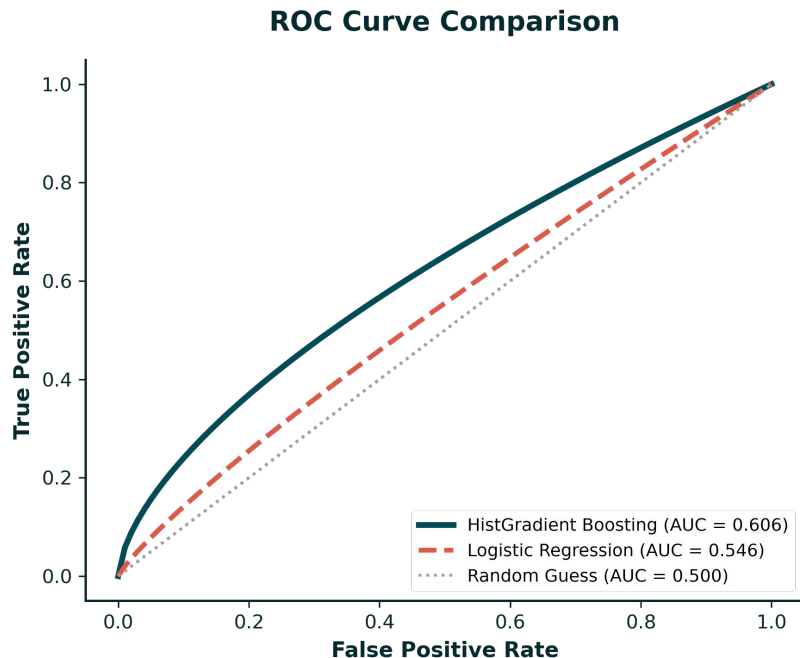
Accuracy 0.54



Linear correlations with **delayed** are practically non-existent across all features, making non-linear tree-based models essential

From Straight Lines to Smart Decision Trees

Hyperparameter tuning demonstrates clear performance gains over the linear baseline



02

HISTGRADIENT BOOSTING



ROC-AUC

0.61



F1 Score

0.64



Accuracy

0.58

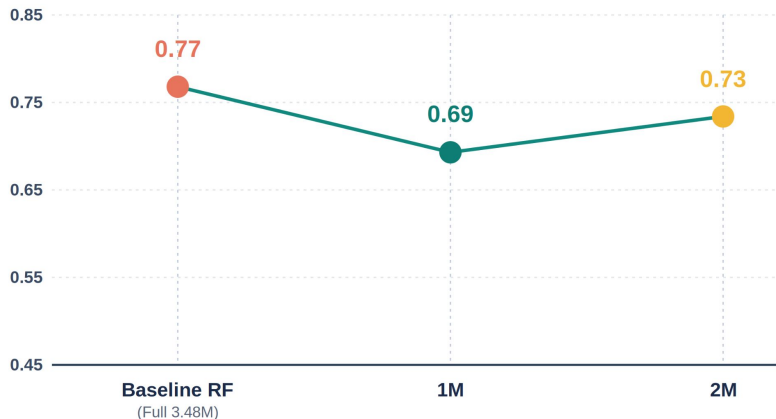


Hyperparameter tuning improves performance over the linear baseline but overall accuracy remains **weak**

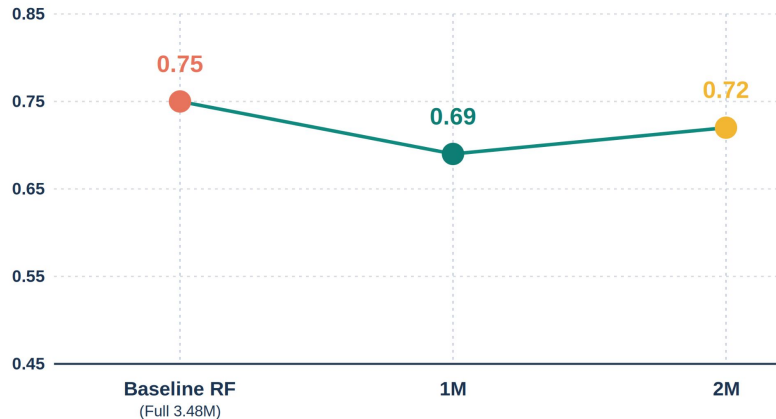
More Data, More Signal

Performance dips with limited data, then recovers as training grows

ROC-AUC (Random Forest)



F1 score - Delayed



BASELINE RANDOM FOREST

Precision (delayed) 0.81

Recall (delayed) 0.69



RANDOM FOREST - 1 MILLION

Precision (delayed) 0.76

Recall (delayed) 0.63



RANDOM FOREST - 2 MILLION

Precision (delayed) 0.79

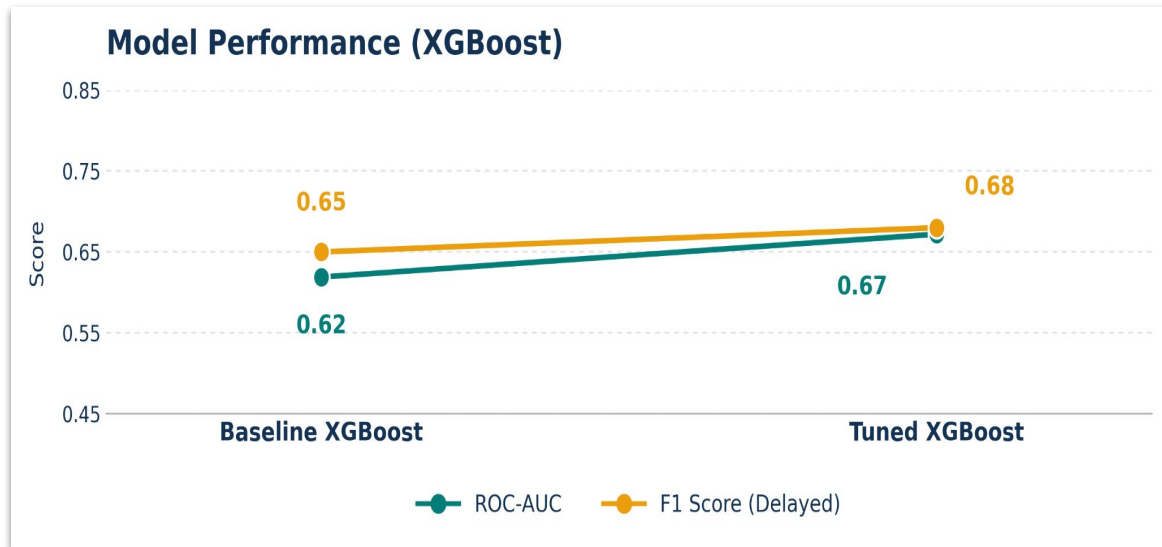
Recall (delayed) 0.66



Random Forest Performance consistently **improved** as training data increased

Smarter Tuning, Stronger Signal

Hyperparameter tuning improves XGBoost across both evaluation metrics



BASELINE XGBOOST	
Precision (delayed)	0.71
Recall (delayed)	0.61

TUNED XGBOOST	
Precision (delayed)	0.74
Recall (delayed)	0.63



Tuning improved the balance between precision and recall for delayed trains.

Key Findings

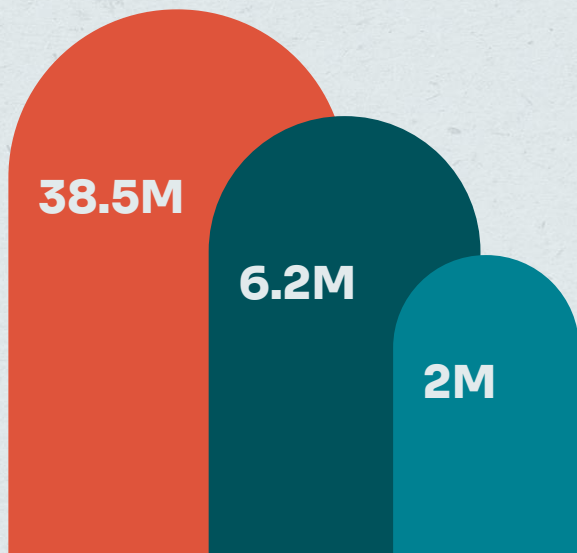
	HIST GRADIENT	RANDOM FOREST	XGBOOST
ROC-AUC	0.61	0.77	0.67
F1 Score	0.64	0.72	0.68
Accuracy	0.58	0.69	0.62

Best Overall Performer



Random Forest led across all metrics
None of the models surpassed the 0.7 overall accuracy barrier

Challenges and Learning



Full Historical Scope

Started with the full 38.5M record dataset, which was too large to process on our machines.

Focus on 2025 data

Filtered to 6.2M records from 2025 to focus on recent data and remove old, outdated trends

Practical Modeling Sample

Sampled down to 2M records to stop computers from crashing and speed up model testing.



Challenges and Learning



Selecting Features

We started with 125 raw columns and ended up with 14 relevant features



Resampling Imbalanced Data

Chose undersampling to manage compute bottlenecks on our massive dataset.



Insights / Conclusion

01

Useful signal, but weak alone

Weather and schedule data contain predictive signal, but no single feature strongly explains delay

02

Complex patterns improve prediction

Tree-based models capture nonlinear relationships and outperform linear models (e.g., Random Forest: ROC-AUC 0.77)

03

Performance has limits

More training data improves performance, but delays remain only partially predictable using weather and schedule data.

CONCLUSION

Machine learning can predict train delays to some extent — but weather and schedule data alone are not enough.

Possible Next Steps

Operational Data



Add train maintenance logs



Live alerts for broken track signals or switch failures.



Platform crowding and boarding delays at busy stations.

Chain Reactions



Model how a single late train creates chain-reaction delays



Add historical delay tracking



The background of the slide features a stylized mountain range in a muted teal color. The mountains are composed of various geometric shapes, creating a modern, minimalist aesthetic. The sky is a light, textured grey, and the foreground is a solid light grey. The overall design is clean and professional.

Thanks!

Team: Amanda Joseph and Winifred Paul